INVERTIS UNIVERSITY, BAREILLY

B.Tech CSE – Third Year | DBMS & SQL Lab (BCS552)

EXPERIMENT NO. 8

1. Experiment Title:

Grouping Rows and Using HAVING

2. Aim:

Group sales by region, count records, calculate regional totals, and filter groups using HAVING.

3. Software / Tools Required:

SQL database system (such as MySQL, PostgreSQL, or SQLite where supported), SQL editor/IDE, and a computer.

4. Theory:

GROUP BY combines rows with the same value into groups. Aggregate functions can calculate values for each group. HAVING filters groups after aggregation, while WHERE filters individual rows before grouping.

5. Algorithm / Procedure:

1. Create shop_sale and insert the supplied records.
2. Group by region and count sales rows in each region.
3. Group by region and calculate the sum of amount.
4. Use HAVING to show only regions whose total is greater than 20000.

6. Program / Queries / Commands:

```
CREATE TABLE shop_sale (
    salesperson VARCHAR(20),
    region VARCHAR(20),
    amount INT
);

INSERT INTO shop_sale (salesperson, region, amount)
VALUES
    ('Deepak', 'North', 12000),
    ('Pooja', 'South', 8500),
    ('Deepak', 'South', 9500),
    ('Lata', 'North', 11000);

-- Task 1
SELECT region, COUNT(*) AS num_sales
FROM shop_sale
GROUP BY region
ORDER BY region;

-- Task 2
SELECT region, SUM(amount) AS total
FROM shop_sale
GROUP BY region
ORDER BY region;

-- Task 3
SELECT region, SUM(amount) AS total
FROM shop_sale
GROUP BY region
HAVING SUM(amount) > 20000
ORDER BY region;
```

7. Input:

North amounts: 12000 and 11000. South amounts: 8500 and 9500.

8. Output:

Expected output (column spacing and display format may differ in your SQL tool):

Task 1: region	num_sales
North	2
South	2
Task 2: region	total
North	23000
South	18000
Task 3: region	total
North	23000

9. Result:

The SQL statements for Experiment 8 were prepared to perform the stated tasks. On executing the setup and queries successfully, the expected results shown above should be obtained.

10. Viva-Voce Questions:

1. What is GROUP BY used for?
2. What is the purpose of HAVING?
3. How is HAVING different from WHERE?
4. Can aggregate functions be used with GROUP BY?